Construction Chemicals

SUBMITTAL DOCUMENT

O-Seal PVC

PVC Waterproofing Membrane
1.5 / 2 / 3 mm



Root Resistant	Welded Seams	Elongation	Signal Layer
Green roofs	Hot air	≥ 300 %	Two colours

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

O-Seal PVC — PVC Waterproofing Membrane 1.5 / 2 / 3 mm

Project Name		**Document No.**	OMX-MS-OSEALPVC
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Seal PVC	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of O-Seal PVC on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers substrate preparation, laying, hot-air welding, seam testing and protection of O-Seal PVC waterproofing membrane.

- Substrate inspection and preparation.
- Installation of protection geotextile.
- Laying and welding of the membrane.
- Seam testing.
- Protection and handover.

3. Definitions

Term	Meaning
Hot-air welding	Joining PVC sheets by melting the overlap with hot air and pressing.
Double-track weld	Two parallel welds with an air channel for pressure testing.
Signal layer	Contrasting colour layer that shows damage.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.

Role	Responsibility
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
O-Seal PVC	PVC membrane rolls 2.2 m × 20 m
Geomat	Protection geotextile
Fixings	Termination bars, PVC-coated sheet profiles

5.1 Key product data

Property	Typical Value
Material	Flexible PVC
Thickness	1.5 / 2 / 3 mm
Tensile strength	≥ 15 – 16 N/mm²
Elongation	≥ 300 %
Tear resistance	≥ 400 – 500 N
Fire class	E (B1)
Cold flexibility	-20 °C to -30 °C
VOC	Free

6. Tools & Equipment

- Automatic and manual hot-air welders
- Silicone rollers
- Seam probe
- Air-pressure test kit
- Knives
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Safety glasses
- Safety shoes

7. Storage & Handling

- Store rolls horizontally in a dry, shaded place, away from heat and sharp objects.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Store rolls and cartons flat, off the ground, protected from direct sunlight, heat, oil and mechanical damage.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Check concrete strength (≥ 25 MPa) and surface smoothness.
2. Remove sharp edges, debris and protrusions.
3. Lay Geomat protection geotextile where specified.
4. Obtain QA/QC approval of the substrate before laying.

10. Trial Welding

1. Check membrane thickness and roll labels.
2. Carry out trial welds at the start of each shift.
3. Peel-test the trial weld and adjust temperature and speed.

11. Laying & Welding

1. Unroll the membrane and allow it to relax.
2. Overlap adjacent sheets by 5 – 10 cm.
3. Weld overlaps with an automatic welder; use manual welders at details.
4. Use double-track welds in critical areas.

5. Fix terminations with termination bars.
6. Record welds and test results on a seam log.

12. Testing & Protection

- Probe all single welds; air-pressure test double-track welds.
- Repair defects with patches welded around.
- Cover with protection layer (Geomat, screed or board) before backfilling.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before laying	≥ 25 MPa, smooth	QA/QC
Overlaps	During laying	5 – 10 cm	QA/QC
Seams	After welding	100 % tested, no leaks	QA/QC + Consultant
Protection	Before backfill	Installed	Site Engineer

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Thickness, roll condition	Each delivery	Material inspection report
Substrate	Smooth, clean	Each area	Inspection request (hold point)
Trial weld	Peel test	Each shift	Weld log
Seams	Probe / air-pressure test	All seams	Seam test record
Protection	Installed before backfill	Each area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Weak weld	Wrong temperature or dirty overlap	Clean and re-weld or patch
Punctures	Sharp objects	Patch with welded piece
Wrinkles at seams	Membrane not relaxed	Cut and patch

15. Health, Safety & Environment

Hazard	Control Measures
Burns from welders	Trained operators; gloves
Fumes	Correct temperature; ventilation
Cuts	Cut-resistant gloves

Hazard	Control Measures
Work in excavations	Shoring and safe access

- Hot-air welders reach high temperatures; avoid burns.
- Avoid inhaling fumes from overheated PVC.
- Use cut-resistant gloves when cutting.
- Burns: cool with water and seek medical advice.
- Inhalation of fumes: move to fresh air.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean welding nozzles with a wire brush.

17. Limitations

- Do not weld in rain or on wet overlaps.
- Overlaps 5 – 10 cm.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Substrate approved	☐ Yes ☐ No	
Geotextile laid	☐ Yes ☐ No	
Trial weld passed	☐ Yes ☐ No	
Membrane laid with overlaps	☐ Yes ☐ No	
Seams welded	☐ Yes ☐ No	
Seams tested	☐ Yes ☐ No	
Details terminated	☐ Yes ☐ No	
Protection installed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

O-Seal PVC — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Waterproofing

O-Seal PVC

PVC Waterproofing Membrane
1.5 / 2 / 3 mm



Product Description

O-Seal PVC is a flexible PVC waterproofing membrane available in 1.5, 2 and 3 mm. It resists tearing, roots and chemicals, contains no VOC, and is installed with hot-air welded seams. Two-colour layers make damage easy to detect.

Uses

- Water tanks and reservoirs.
- Channels and tunnels.
- Foundations and basements.
- Industrial ponds.
- Roof gardens and green roofs.

Advantages

- Root and tear resistant.
- Chemical resistant.
- Hot-air welded seams.
- Two-colour signal layer for inspection.
- Elongation ≥ 300 %.

Application

1. **Surface Preparation**
   Concrete ≥ 25 MPa, smooth, clean; install protection geotextile (Geomat).
2. **Laying**
   Unroll the membrane with 5 – 10 cm overlaps.
3. **Welding**
   Weld overlaps with hot air; use double-track welds in critical areas and test seams.

Consumption

approx. 1.1 m² per m²
including 5 – 10 cm overlaps
Add details and upstands as per drawings.

Packaging

Rolls 2.2 m × 20 m; thickness 1.5, 2 or 3 mm.

Storage & Shelf Life

Store rolls horizontally in a dry, shaded place, away from heat and sharp objects.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Material	Flexible PVC	—
Thickness	1.5 / 2 / 3 mm	—
Tensile strength	≥ 15 – 16 N/mm²	—
Elongation	≥ 300 %	—
Tear resistance	≥ 400 – 500 N	—
Fire class	E (B1)	—
Cold flexibility	-20 °C to -30 °C	—
VOC	Free	—
Roll size	2.2 m × 20 m	—

Consumption

Item	Requirement
Overlap	5 – 10 cm
Critical areas	Double-track weld with air channel
Substrate	Concrete ≥ 25 MPa, smooth

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Substrate smooth, clean and free of sharp edges.
- Concrete compressive strength ≥ 25 MPa.
- Weld only clean, dry overlaps.
- Do not weld in rain.

Application Notes

- Use Geomat geotextile as a protection layer under and over the membrane where specified.
- Weld trial seams each day to set temperature and speed.
- Test seams by probing; test double-track seams by air pressure.
- Protect the finished membrane before backfilling or screeding.

Health & Safety

- Hot-air welders reach high temperatures; avoid burns.
- Avoid inhaling fumes from overheated PVC.
- Use cut-resistant gloves when cutting.

First Aid & Cleaning

- Burns: cool with water and seek medical advice.
- Inhalation of fumes: move to fresh air.
- Cuts: clean and dress the wound.
- Clean welding nozzles with a wire brush.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

O-Seal PVC — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

O-Seal PVC — Submittal Document

To be attached: approvals and certificates, where available.